

PARTICLE PHYSICS

BY

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RESEARCH AND THEORY

EXACTNESS, DISCRIMINATION, AND THE ACCUMULATION OF MATTER

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*For Megan, my loving wife, the only person that really understands
what I mean when I speak.*

*For Autumn and Opal, without you this thought would have never
occurred.*

To everyone else, I wish you well.

ABSTRACT

The subject: what exact arithmetic lets a reading keep. The orior engine reaches a symbol through equality alone, and a point either carries the same symbol as another or it does not, with no order and no distance imposed from the reading end. On integers that equality is exact, and an exact reading never merges two states that differ. This research paper reads the atom that way: an element is the electrons of its neutral atom, each an integer quantum state, and the whole periodic table is one accumulation of them.

Pauli exclusion is an accumulation, and the reading keeps it. Two fermions may not share a state. Electrons accumulate into successive states instead of piling into the lowest, and that stacking is the shell structure. The exclusion principle appears here as a count: the same domain-blind primitive that finds two elements on one crystallographic site finds two electrons in one state, and over all 118 elements it finds none. Coarsen the state and distinct electrons agglomerate into one cell; the fullest cell at each coarsening is the Pauli capacity for that level, read off the accumulation with nothing put in.

The periodic law is the recurrence, measured against a drawn null. An element's group signature recurs down the table, and the recurrence is measured by an equality match rate read against two backgrounds: a shuffle that keeps the counts and deletes the order, and a filling that keeps no exclusion. The recurrence survives neither. It is in the order Pauli fills the shells and not in the counts, and removing the exclusion leaves a ladder with no table. The row lengths, 8, 8, 18, 18, 32, 32, are read off the shell closures as the differences between them.

What is measured and what is deferred. The table pipeline runs on the ideal filling, Madelung order under Hund's rule, and its figures are deterministic. A later chapter carries the Bohr trajectory in exact rationals, where a radius is n^2/Z and an energy is $-Z^2/n^2$ with nothing rounded, and a whole spectral series reduces to line ratios that carry no measured constant at all. The oracle chapter holds that ideal filling against the measured ground states from the NIST database, covering all 118 elements, the measured configurations through element 108 and the predicted ones above it, and reads off the aufbau exceptions where a real atom fills against the order. What is not run is the comparison of an absolute Bohr line to a measured wavelength, which needs the recommended constants and is named, not fetched. A final reading chapter takes the Standard Model particles, put in from the particle data, for the structure their exact charges keep. The claims that rest on measurement say so, and the claims that are exact arithmetic say that instead.

CONTENTS

1	EXACTNESS AND INFINITE DISCRIMINATION	1
1.1	The engine reads by equality	1
1.2	A coarse reading imposes a quantum	1
1.3	Pauli as a count	2
1.4	The partition is the quantum numbers	2
2	ACCUMULATION AND THE PERIODIC LAW	3
2.1	The ideal filling	3
2.2	Exclusion is an accumulation	3
2.3	The recurrence, against a drawn null	3
2.4	The row lengths, off the difference set	4
3	TRAJECTORIES IN EXACT ARITHMETIC	5
3.1	The orbit is a rational	5
3.2	One hub, propagated exactly	5
3.3	The ratios carry no constant	6
3.4	What the oracle needs, and what the model is	6
4	THE PARTICLES, AND THE STRUCTURE THE EXACT NUMBERS KEEP	7
4.1	Charge in thirds, and no mass	7
4.2	The generations recur	7
4.3	The charge sum vanishes	8
4.4	Why a quark is never alone	8
4.5	The hadrons the quarks make	8
4.6	The rarest is the last found	8
5	SOURCES	9
	BIBLIOGRAPHY	11

EXACTNESS AND INFINITE DISCRIMINATION

Every figure in this chapter is computed on the ideal filling and is deterministic; the runs that produced them are named at each claim. The ideal filling is Madelung order under Hund's rule, stated in full in the next chapter. What is exact here is the arithmetic, and the figures measure what it keeps.

1.1 THE ENGINE READS BY EQUALITY

The orior engine reaches a symbol through one operation. Its field interface answers, for two positions in a corpus, whether they carry the same symbol: no order on the symbols, no distance between them, and no alphabet they are drawn from. A measure built on that interface can read an element, a letter or a chess move without knowing which, because all it ever asks is whether two things are the same thing.

On integers that question is exact. Two integer quantum states are the same state or they are not, and the reading never has to set a threshold for how close is close enough. That is the property this research paper rests on, and it is worth stating plainly before any atom is read: where the symbols are exact, the discrimination is unbounded. Nothing the source distinguished is lost, because nothing was rounded into anything else on the way in.

1.2 A COARSE READING IMPOSES A QUANTUM

The ingestion primitive keeps every digit its source wrote, and its header states the discipline it holds to: no quantum is imposed from the reading end. A quantum is a decision about the smallest difference worth telling apart, and the crystallography reader once made one silently, putting every deposited site on a grid of a quarter of an angstrom. That the recovered cell edge carried the grid and not the deposit. The exact reading of the same cells answers the published edge with an error of zero, and the difference between the two readings is the grid.

An atom has the same failure available. An electron sits at a quantum state of four numbers, the principal, azimuthal, magnetic and spin numbers, and a reading that keeps only the first two, the subshell, has dropped the two that tell the electrons of one subshell apart. The choice is the voxel again, in a different domain.

1.3 PAULI AS A COUNT

The representation is the electron set of the neutral atom, one point per electron at an integer state, built in `examples/particle_physics/1_represent/an_atom_as_exact_shells.py`. Two electrons at one state is a Pauli violation, and the same domain-blind primitive that finds two elements on one crystallographic site finds it: `exact.contested` returns the positions carrying more than one value. Over all 118 elements the exact reading places Z electrons at Z distinct states and no state is contested. Exclusion holds for every element the ledger names.

The check is proven by breaking its subject. A filling that placed two electrons at one state gives `exact.placed`, a dictionary, one entry where the count expects two. The distinct-state count falls below Z and the run reports it. A correct fill of two electrons keeps two states; a doubled state keeps one. The count separates them, and the labels never disagree, because both electrons carry the same subshell. The loss under a coarse reading is a count and only a count, and a check that compares values reports nothing: across the table the subshell reading collapses 5645 electron states and a value-comparing check flags none of them.

1.4 THE PARTITION IS THE QUANTUM NUMBERS

The partition of an atom is which of the four numbers the reading keeps, and `examples/particle_physics/2_partition/what_a_quantum_number_costs.py` sweeps them. Keep all four and every electron stands at its own state. Drop the spin and the two electrons of an orbital fall to one cell. Drop the magnetic number and a subshell falls together. Drop the principal number and the atom is one cell.

The fullest one cell ever gets at each partition is the capacity Pauli allows there, and it is read off the accumulation and not put in: one electron per spin orbital, two per orbital, fourteen at the f subshell, thirty two at a full $n = 4$ shell. Only the full partition keeps every element's electrons apart. Every coarser one agglomerates, and hydrogen, with one electron to fold, is the only element that survives every partition. Exact arithmetic is the full partition, and the reading that never agglomerates two states that differ is the reading this whole engine is built to be.

ACCUMULATION AND THE PERIODIC LAW

The figures here are measured on the ideal filling and are deterministic. The ideal filling is defined first, because every later claim is a claim about it and not about a measured atom.

2.1 THE IDEAL FILLING

The ledger at `src/engine/python/representation/atom/element.py` builds an element's electrons in Madelung order, lowest principal plus azimuthal first, and Hund's rule inside a subshell, every magnetic orbital taking a spin-up electron before any takes spin-down. The order fills 1s through 7p, and its capacities run to 118 exactly. It names every element from hydrogen to oganesson and no further.

This is the necessary condition, the filling the periodic law would follow if nothing competed with it. Real atoms deviate: chromium, copper and others fill against the order, and those measured configurations belong to the oracle stage of the next chapter. The represent field carries the ideal, and it says so. That no reader takes the ideal for a measurement.

2.2 EXCLUSION IS AN ACCUMULATION

Two fermions may not occupy one state. An electron added to an atom therefore cannot join the electrons already in the lowest state; it takes the next state the exclusion leaves open, and the one after that takes the next again. Electrons accumulate into successive states, and the stack of occupied states is the shell structure. Exclusion is not a bookkeeping rule laid over the atom. It is the reason the atom has structure at all.

That accumulation is faithful here for a reason the first chapter gave. A reading that rounded two close states together would lose an electron from the stack and shorten a shell. The exact reading never does, because its discrimination is unbounded, and an accumulation of fermions stays an accumulation and never collapses. Exactness lets the engine hold the stack, and exclusion makes the stack tall.

2.3 THE RECURRENCE, AGAINST A DRAWN NULL

The differentiating electron of an element is the last one the order adds, and its group signature is its azimuthal type and its subshell's population with the principal number dropped. Dropping the principal number leaves sodium and lithium reading the same signature, and that recurrence

down the table is the periodic law. It is measured by an equality match rate, how often a signature equals the one a fixed number of elements ahead, in `examples/particle_physics/3_reference/what_exclusion_builds.py`.

A match rate means nothing without the rate a shuffle of the same signatures reaches. Two backgrounds are drawn, each deleting a property from the data. The first keeps the census and deletes the order: a uniform permutation of the same signatures, which fixes every count exactly. Its floor is the real floor, and the match rate at the row lengths falls to it. The recurrence is in the order the shells fill and not in the counts. The second deletes the exclusion: every electron drops to 1s, an element becomes a count of electrons in one state, every signature is distinct, and the match rate and its floor are both zero. Without exclusion there is a ladder of counts and no table. The real accumulation is the positive control, and it is the only arm that carries the recurrence.

2.4 THE ROW LENGTHS, OFF THE DIFFERENCE SET

The recurrence does not sit at one lag. The match rate stands above its floor at 32, at 18 and at 8, and at their sums, because the rows of the table are 2, 8, 8, 18, 18, 32, 32 and not one length. A reader reporting a single period would be wrong for most of the table, the harmonic trap the periodicity primitive records from the other side. What is one number per row is the gap between shell closures, and `examples/particle_physics/4_measure/shell_closures_from_the_difference_set.py` reads them off the signatures: a closure is a filled p subshell, with helium closing the first row.

closure at Z	2	10	18	36	54	86	118
row length		8	8	18	18	32	32

The closures are the noble gases, and the gaps between them are the row lengths, with no row told to the reader. This is the difference-set reading of the crystallography chapter moved to one dimension: a period is a difference between two things that agree. The differences are the complete candidate set.

TRAJECTORIES IN EXACT ARITHMETIC

The reproduction in this chapter is exact and the comparison to measurement is not run. The radii, energies and line ratios below are exact rationals, facts of the Bohr model with no digit approximated. Whether a real atom matches them is the oracle stage, and it is named at the end of the chapter, not carried out.

3.1 THE ORBIT IS A RATIONAL

A hydrogen-like ion is one electron on a nucleus of charge Z . Bohr quantizes the angular momentum at an integer multiple of $\hbar$, and the orbit that follows has radius n^2/Z in units of the Bohr radius and energy $-Z^2/n^2$ in units of the Rydberg. Both are exact rationals, an integer over an integer, built in `examples/particle_physics/1_represent/an_orbit_as_exact_rationals.py` with integer arithmetic done by hand, the host stand-in for the fixed-width limb integers the device arm carries in `no_rounding`.

The quantum number n is an exact integer. Two neighboring orbits differ by

$$\frac{(n+1)^2}{Z} - \frac{n^2}{Z} = \frac{2n+1}{Z},$$

exactly, and no two orbits ever fall on one radius, however large n grows. That is the property a floating point orbit loses: past some n the gap between two orbits falls below the last bit, and the two large orbits merge. The exact reading keeps them apart forever, the same infinite discrimination the shells were read with, now carrying a trajectory instead of a state.

3.2 ONE HUB, PROPAGATED EXACTLY

Every energy level of every hydrogen-like ion is $E(n, Z) = -\text{Ry } Z^2/n^2$, and every radius is $r(n, Z) = a_0 n^2/Z$. One physical quantity sets each, the Rydberg energy and the Bohr radius, and the whole grid of levels and radii follows from it by an exact rational identity. Compute the hub once and the ladder is fixed to the precision of the hub, with nothing added by the arithmetic that spreads it.

The split this draws is the split a precision program needs. The ratios n^2/Z and $1/n^2$, the integer quantum numbers, the capacities and the closures are mathematical: their uncertainty is exactly zero and there is nothing in them to measure better. The Rydberg energy, the Bohr radius, the fine-structure constant and the electron-to-proton mass ratio are physical: they are measured, their uncertainty is irreducible, and they are the only floor the exact quantities stand on.

3.3 THE RATIOS CARRY NO CONSTANT

Within one spectral series the hub cancels. A transition from n_2 to n_1 has photon energy proportional to $1/n_1^2 - 1/n_2^2$, and the ratio of two lines of one series is a pure rational with no Rydberg and no meter left in it. The Balmer series, $n_1 = 2$, read as wavelength ratios to its first line:

line	upper n	ratio	decimal
H α	3	1	1.00000
H β	4	20/27	0.74074
H γ	5	125/189	0.66138
H δ	6	5/8	0.62500

These ratios are predictions with no measurement in them, infinitely precise, and a whole ladder of them stands on no physical constant at all. A measured ratio meets them at the oracle, and the meeting needs no constant either, and the comparison stays clean.

3.4 WHAT THE ORACLE NEEDS, AND WHAT THE MODEL IS

The absolute size of an orbit in meters, and any comparison of an absolute line to a measured wavelength, needs the Rydberg constant and the Bohr radius, which are measured and cited, not derived. That comparison is the oracle stage. It is not run here, because it needs the measured configurations and the measured spectra, and those are not fetched. Naming a source is not reading it, and no measured wavelength is quoted in this research paper.

The floor on a real spectral line is not the arithmetic, which stays exact, but two things above it. One is the measured constants. The other, larger by far, is the model: the Bohr picture omits fine structure, the Lamb shift and every later correction. The model error dominates the constants' uncertainty long before the arithmetic enters. The exact rationals are the model reproduced without loss. How far the model itself stands from a measured atom is the question the oracle exists to answer, and it is left open here.

THE PARTICLES, AND THE STRUCTURE THE EXACT NUMBERS KEEP

The particles here are put in, and the structure is read out. The confirmed content of the Standard Model, six quarks, six leptons and five force carriers, comes from the Particle Data Group at CERN, and nothing in this chapter derives a particle from first principles. What the chapter shows is how the exact quantum numbers organize themselves once the masses are set aside, and the reading that does it is the same equality the shells were read with.

4.1 CHARGE IN THIRDS, AND NO MASS

Electric charge is quantized in thirds of the electron charge. An up quark is $+2$, a down quark is -1 , a charged lepton is -3 , a neutrino is 0 and a W is ± 3 , all in units of a third, and carried that way each is an exact integer. Spin is doubled in the same manner, a fermion reading 1 and a gauge boson 2 , and baryon number is carried in thirds. The ledger at `src/engine/python/representation/particle/standard_model.py` holds all of it as integers.

No mass enters. A mass is a measured real number, and reading structure from it would mean choosing a tolerance, a bound on how close two masses must be to count as one. This work refuses a bound everywhere, and it refuses one here: every question the chapter asks is answered by equality or by an exact sum, never by a threshold. Two particles carry the same charge or they do not; a generation's charges sum to zero or they do not.

4.2 THE GENERATIONS RECUR

Strip a fermion's generation and its mass, and its signature remains: its charge, its spin, its color count, its baryon and lepton numbers. The up, charm and top quarks share one signature. The three charged leptons share another. Read over the three generations, the signature sets are equal. The generation is one structure seen three times, the way a periodic group was one signature seen down a column in the second chapter. The only difference between an electron and a muon this reading can see is a mass, and the reading does not look at it.

That recurrence is the periodic table's lesson at a lower level. There the valence signature repeated down a column and the instrument read the group without being told chemistry. Here the fermion signature repeats across a generation and the instrument reads the family without being told a mass.

4.3 THE CHARGE SUM VANISHES

Sum the electric charge of one generation, counting each quark once for each of its three colors:

$$3 \cdot (+2) + 3 \cdot (-1) + (-3) + 0 = 0,$$

in thirds, and the same for the second generation and the third. The sum is exactly zero. That is the anomaly-free condition the theory needs to be consistent, and it is the reason an atom is neutral: the three quarks of a proton carry +3 thirds and the electron carries −3, and they balance to the digit. The thread from the particle ledger to the atom ledger of the earlier chapters is this exact zero. An atom is neutral because a generation's charges cancel, and they cancel exactly because charge is an integer in thirds and the color count is exactly three.

4.4 WHY A QUARK IS NEVER ALONE

A charge that divides by three is an integer charge. The leptons and the bosons all carry one, and they are the particles seen free. The six quarks do not: their charges are +2 and −1 in thirds, which no count of three divides. A quark alone would carry a fractional charge, and none has been seen. Only combinations whose thirds add to a multiple of three run free, the proton at +3 thirds and the neutron at 0, and the exact arithmetic of the thirds is all of it. Confinement has more to it than charge, but the charge alone already forbids a lone quark, and it forbids it by divisibility and not by any bound.

4.5 THE HADRONS THE QUARKS MAKE

The combinations that run free are read in `examples/particle_physics/1_represent/hadrons_from_quarks.py`. A baryon is three quarks and a meson is a quark and an antiquark, and over the light quarks u, d and s the ten three-quark combinations carry the charges of the baryon decuplet, the proton at +1, the neutron at 0 and the omega at −1 among them, while the nine quark-antiquark combinations carry the pion and kaon charges. Every one of the nineteen carries an integer charge, a multiple of three thirds, and a single quark never does. The observed hadron charges are the exact integer sums of their quark charges, put in from the particle data and added up here, and the proton at +1 balancing the electron is again the neutrality of the atom, read now from the quark side. Nothing is derived about why the quarks bind or what a hadron weighs; the charge is a sum of integers.

4.6 THE RAREST IS THE LAST FOUND

The census ranks the quantum numbers by magnitude, the total less a count, the rarest first, the same order the sift kernel probes a field in. Over the eighteen particles the spins fall 1/2 twelve times, 1 five times and 0 once. The scalar is rarest, and a sift over the particles reaches the Higgs first, the particle the collider confirmed last. The reading that ranks it first knows nothing of when it was found; it knows only that a spin of zero occurs once.

SOURCES

Each source is listed with what this research paper takes from it. The read sources are cited by path or by name. The oracle's measured tables are now fetched and read; one database it will need for the absolute spectral scale is named at the end and was not retrieved. Nothing is quoted from it.

THE ENGINE AND THE READING

The engine, six parts. The construction this research paper reads with, represent, partition, reference, measure, sift and oracle, and the rule that a symbol is reached through equality alone. `src/engine/python/README.md` and `src/engine/c/engine/orior.h`.

Exact ingestion. The primitive that keeps every digit and imposes no quantum from the reading end, and placed and contested, the domain-blind readings the Pauli count is built on. `src/engine/python/representation/exact.py`.

The element ledger. One transcribed table of atomic number, symbol and the Madelung filling order, read by this research paper and by the chemistry subject alike. `src/engine/python/representation/atom/element.py`.

The particle ledger. The confirmed Standard Model particles as exact integer quantum numbers, charge in thirds and spin doubled, with no mass. `src/engine/python/representation/particle/standard_model.py`.

The equality match rate. The measure the recurrence is read with, and the coincidence floor a shuffle reaches. `src/engine/python/measure/sequence_match.py`.

The runs. Every figure in this research paper is produced by a script under `examples/particle_physics/`: the atom as exact shells, the orbit as exact rationals and the particle ledger in `1_represent`, the quantum-number partition in `2_partition`, the two drawn nulls in `3_reference`, the shell closures in `4_measure`, and the ideal against the measured ground states in `6_oracle`. The subject's `README.md` carries the figures beside the scripts that print them.

Note, 26 September. At orior 1948ae1 the engine header above is `src/engine/nbody/orior/orior.h`, moved from `src/engine/c/engine/` by commit bdaed61 on 24 September. Every `src/engine/python/` file listed above was deleted by commit 74601c1 on 26 September, and none is in a file at the pin. All 11 scripts under `examples/particle_physics/` import from that tree. Every figure in this research paper was produced before the deletion. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before

the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted is at the pin again; nothing here was rerun against them.

REFERENCE DATA

Element symbols and atomic numbers. The IUPAC table, canonical reference that names the elements 1 to 118. It is standard and carries no single locator; the ledger transcribes it and states so.

The Madelung filling order and Hund's rule. Standard atomic structure. The order fills 1s through 7p and its capacities close on 118, which the ledger checks and does not assert.

THE MEASURED DATA

NIST Atomic Spectra Database. [1] The measured ground-state configurations, fetched by `maint/data/fetch/fetch_nist_ground_states.py` and read by the oracle. It carries neutral atoms through element 108 and holds the aufbau exceptions the ideal filling does not.

The predicted superheavy configurations. Elements 109 to 118 have no measured neutral atom. The oracle takes the calculated configurations, marked predicted and kept apart from the measured ones. Source: Wikipedia, Electron configurations of the elements (data page) [3], which compiles the relativistic predictions past hassium.

The Particle Data Group, at CERN. [2] The confirmed particle content, three generations and the Higgs, and the measured masses, which the particle ledger records and this reading sets aside. Read through the Standard Model overview that compiles it.

NAMED, NOT YET READ

CODATA internationally recommended values. The Rydberg constant, the Bohr radius and the fine-structure constant, the measured hubs that set the absolute scale the exact rationals stand on. The trajectory chapter's absolute comparison needs them and does not run. They are named and not quoted.

BIBLIOGRAPHY

Every measurement in this work is built on somebody's published result, and the first purpose of this chapter is that those people are credited by name. A surname in a footnote is not a citation. The second purpose is that the credit is checkable: each source is registered in `orior_citations`, where the copy sits itself, and this bibliography is generated from that registry and pinned to one commit of it. A reference therefore resolves not to a title but to exact bytes.

Credit here does not depend on agreement. Where this work reproduces a published result it says so, and where it fails to reproduce one it says that too, under the same names, because a refutation rests on the original as much as a confirmation does.

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A reader who disagrees with a number in this research paper can ask which copy it was checked against, and the hash answers exactly. Two copies of a paper are rarely the same bytes.

- [1] National Institute of Standards and Technology. *NIST Atomic Spectra Database*. <https://physics.nist.gov/cgi-bin/ASD/ie.pl>.
- [2] Particle Data Group.
- [3] Wikipedia, Electron configurations of the elements (data page). *Electron configurations of the elements (data page)*. [https://en.wikipedia.org/wiki/Electron_configurations_of_the_elements_\(data_page\)](https://en.wikipedia.org/wiki/Electron_configurations_of_the_elements_(data_page)).